

UDIPTA MONDAL

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PROFESSIONAL SUMMARY

Computer Science graduate focused on building and deploying production AI systems. Researched ML-based predictive modeling for disease risk detection and designed AegisDID (decentralized ZK-ML biometric verification). Builds Generative AI and multi-agent LLM workflows (CrewAI, Gemini API), hybrid RAG pipelines (ChromaDB), real-time computer vision pipelines (YOLOv11, ByteTrack), and Dockerized FastAPI services on cloud endpoints.

EDUCATION

Presidency University Bangladesh

Graduated: December 2025

Computer Science and Engineering | CGPA: 3.10 / 4.00

Dhaka, Bangladesh

Thesis: ML-Based Predictive Modeling for Early Detection of Dengue Disease

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI / ML: Machine Learning, Computer Vision, Generative AI, Agentic AI, LLMs, Prompt Engineering, RAG Pipelines

Frameworks: CrewAI, TensorFlow, Keras, OpenCV, scikit-learn, NumPy, Pandas, MLflow, MediaPipe

Cloud / Deployment: FastAPI, Docker, Git, GitHub, Vercel, Render

Databases: SQL, Vector Databases

Core Concepts: Data Structures & Algorithms, Operating Systems, DBMS, OOP

PROJECTS

AI Tutor Platform: Multimodal Adaptive Learning Engine | LLMs, RAG, ChromaDB, OCR, scikit-learn

- Architected an adaptive AI tutoring system using a hybrid RAG pipeline (ChromaDB + Gemini LLM) for Socratic guidance.
- Built a multimodal grading engine combining OCR for handwritten derivations with ML weak-topic knowledge tracing.

AegisDID: Sybil-Resistant Decentralized Identity | MediaPipe, ZK-ML, Computer Vision, Circom, Groth16

- Designed a privacy-preserving decentralized identity system to prevent duplicate aid claims without a central database.
- Implemented on-chain zero-knowledge proofs (Circom, Groth16) with device-local face matching and one-time nullifiers.

ResearchFlow AI: Autonomous Deep Research Engine | CrewAI, Agentic AI, LLM Orchestration, Multi-Agent, Gemini LLM

- Built a 4-agent autonomous research team with CrewAI and Gemini LLM, orchestrating web search and cited report generation.
- Implemented context-passing task delegation, automated verification passes, and citation-backed markdown report generation.

Falcon AI: Real-Time Border Threat Detection | YOLOv11, ByteTrack, Computer Vision, OpenCV

- Built a real-time border surveillance pipeline using YOLOv11 and ByteTrack to detect push-in/push-out intrusion events.
- Integrated an automated alert dispatch module delivering prioritized threat notifications directly to field personnel.

People Flow Detection & Spatial Heatmap Analytics | YOLOv11, ByteTrack, Computer Vision, OpenCV

- Built an end-to-end pedestrian tracking pipeline with YOLOv11 and ByteTrack to preserve identities across occluded video feeds.
- Implemented bidirectional virtual line counters and 2D Gaussian density heatmaps for real-time spatial congestion monitoring.

Heart Disease Prediction API | Machine Learning, Random Forest, scikit-learn, FastAPI, Docker

- Trained an optimized Random Forest classifier on 13 clinical biomarkers, achieving **83.33% accuracy** and **93.97% ROC-AUC**.
- Deployed the model as a Dockerized async FastAPI service with Pydantic validation and interactive risk scoring.

AWARDS & CERTIFICATIONS

- Blockchain Olympiad Bangladesh | Rank 8 (AI Category, 2025)
- International Data Science Olympiad, Hong Kong | Nominated (2025)
- ICPC Asia Dhaka Regional 2023 | Contestant
- Intra-University Programming Contest | Presidency Univ. — Winner (Season 2), 1st Runner-Up (Season 1); AI Contest — 2nd Runner-Up (2024)